

type.l

```
1  %{
2  #include <stdio.h>
3  #include <stdlib.h>
4  #include "type.tab.h"
5  %}
6
7  %%
8
9  [0-9]+          { return INTEGER; }
10
11  [0-9]+\.[0-9]+  { return FLOAT; }
12
13  [a-zA-Z]+       { return CHAR; }
14
15  [ \t]+          ;
16
17  \n              { return EOL; }
18
19  .               { return yytext[0]; }
20
21  %%
22
23  int yywrap()
24  {
25      return 1;
26  }
27
```

type.y

```
1  %{
2  #include <stdio.h>
3
4  int yylex();
5
6  void yyerror(const char *s)
7  {
8      fprintf(stderr, "Parse error: %s\n", s);
9  }
10 %}
11
12 %token INTEGER FLOAT CHAR EOL
13
14 %%
15
16 program:
17     /* empty */
18     | program line
19     ;
20
21 line:
22     statement EOL
23     ;
24
25 statement:
26     INTEGER
27     {
28         printf("Type: INTEGER\n");
29     }
30     | FLOAT
31     {
32         printf("Type: FLOAT\n");
33     }
34     | CHAR
35     {
36         printf("Type: CHAR/STRING\n");
37     }
38     ;
39
40 %%
41
42 int main()
43 {
44     yyparse();
45     return 0;
46 }
47
```



/c/Users/Afroz/Desktop/EXP6

```
Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/EXP6
$ ./type
123
Type: INTEGER
asd
Type: CHAR/STRING
123.123
Type: FLOAT
asd123
Type: CHAR/STRING
Parse error: syntax error

Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/EXP6
$
```